

Questions with Answer Keys

MathonGo

Q1 - 2024 (01 Feb Shift 1)

A tuning fork resonates with a sonometer wire of length 1 m stretched with a tension of 6 N. When the tension in the wire is changed to 54 N, the same tuning fork produces 12 beats per second with it. The frequency of the tuning fork is _____ Hz.

Q2 - 2024 (27 Jan Shift 2)

A closed organ pipe 150 cm long gives 7 beats per second with an open organ pipe of length 350 cm, both vibrating in fundamental mode. The velocity of sound is _____ m/s.

Q3 - 2024 (30 Jan Shift 1)

In a closed organ pipe, the frequency of fundamental note is 30 Hz. A certain amount of water is now poured in the organ pipe so that the fundamental frequency is increased to 110 Hz. If the organ pipe has a cross-sectional area of 2 cm^2 , the amount of water poured in the organ tube is _____ g. (Take speed of sound in air is 330 m/s)

Q4 - 2024 (30 Jan Shift 2)

A point source is emitting sound waves of Power $16 \times 10^{-8} \text{ W}$ at the origin. The ratio of intensity (magnitude only) at two points located at a distances of 2 m and 4 m from the origin respectively will be _____ Wm^{-2} .

[We modified question statement, In official NTA paper question was incomplete]

Q5 - 2024 (31 Jan Shift 1)

The fundamental frequency of a closed organ pipe is equal to the first overtone frequency of an open organ pipe. If length of the open pipe is 60 cm, the length of the closed pipe will be :

- (1) 60 cm
- (2) 45 cm
- (3) 30 cm
- (4) 15 cm

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Q6 - 2024 (31 Jan Shift 2)

The speed of sound in oxygen at S.T.P. will be approximately:

(Given, $R = 8.3 \text{ JK}^{-1}$, $\gamma = 1.4$)

(1) 310 m/s

(2) 333 m/s

(3) 341 m/s

(4) 325 m/s

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Solutions

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Q1

$$f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

$$f_1 = \frac{1}{2} \sqrt{\frac{6}{\mu}} \quad f_2 = \frac{1}{2} \sqrt{\frac{54}{\mu}}$$

$$\frac{f_1}{f_2} = \frac{1}{3} \quad f_2 - f_1 = 12$$

$$f_1 = 6\text{Hz}$$

Q2



$$f_c = \frac{v}{4\ell_1} \quad f_o = \frac{v}{2\ell_2}$$

$$|f_c - f_o| = 7$$

$$\frac{4 \times 150}{v} - \frac{2 \times 350}{v} = 7$$

$$\frac{600 \text{ cm}}{v} - \frac{700 \text{ cm}}{v} = 7$$

$$\frac{v}{6 \text{ m}} - \frac{v}{7 \text{ m}} = 7$$

$$v \left(\frac{1}{42} \right) = 7$$

$$v = 42 \times 7$$

$$= 294 \text{ m/s}$$

Q3

$$\frac{V}{4\ell_1} = 30 \Rightarrow \ell_1 = \frac{11}{4} m$$

$$\frac{V}{4\ell_2} = 110 \Rightarrow \ell_2 = \frac{3}{4} m$$

$$\Delta\ell = 2m$$

$$\text{Change in volume} = A\Delta\ell = 400 \text{ cm}^3$$

$$M = 400\text{g}; (\because \rho = 1 \text{ g/cm}^3)$$

Q4

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Solutions

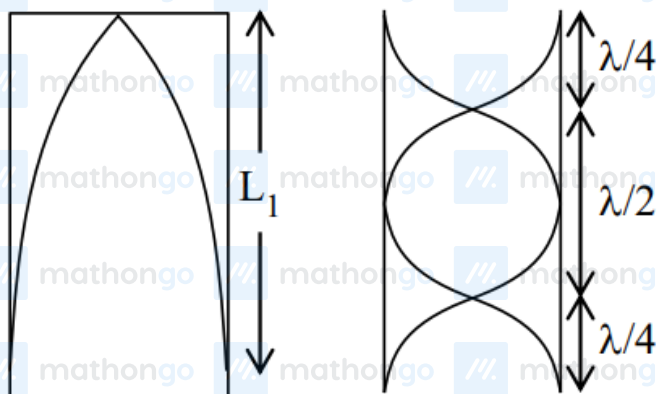
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Intensity = Power/Area

Ratio of intensity will be inverse square ratio of distances

$$\text{Ratio} = \frac{4^2}{2^2} = 4$$

Q5



$$\frac{\lambda}{4} = L_1 \quad 2 \left(\frac{\lambda}{2} \right) = \lambda$$

$$v = f\lambda \quad f_2 = \frac{2v}{\lambda}$$

$$v = f_1 (4 L_1) \quad f_2 = \frac{v}{L_2}$$

$$f_1 = \frac{v}{4 L_1}$$

$$f_1 = f_2$$

$$\frac{v}{4 L_1} = \frac{v}{L_2}$$

$$\Rightarrow L_2 = 4 L_1$$

$$60 = 4 \times L_1$$

$$L_1 = 15 \text{ cm}$$

Q6

$$v = \sqrt{\frac{\gamma RT}{M}} = \sqrt{\frac{1.4 \times 8.3 \times 273}{32 \times 10^{-3}}}$$

$$= 314.8541 \simeq 315 \text{ m/s}$$

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